Separation Units

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Introduction

Importance of Equipment:

- Equipment plays an essential role in chemical engineering processes by turning raw materials into finished products.
- In addition to creating products; equipment contributes to efficiency and cost reduction, quality and safety, along with research and development.
- The wide applicability of process equipment proves why every engineer needs to have a thorough understanding of the theory, capabilities, and limitations of all unit operations they're working with.

Introduction (cont.)

Categories of Separation Equipment:

Distillation Columns – a very common liquid-vapor separation process that's carried out in industry. Can be a batch or continuous process. This unit operation works by applying and removing heat to exploit the differences in relative volatility between components. More volatile (lower b.p.) components rise as vapor, while less volatile components (higher b.p.) settle to the bottom as liquid.

Separators - deal with mechanical operations that separate components based on their physical properties such as density, particle size, or phase using physical forces such as gravity, centrifugal force, or momentum. They don't rely on chemical changes or temperature-driven phase changes.

When to Use & Column Choices

Use distillation when the volatility varies ($\alpha \gtrsim 1.1$) with sharp splits and high throughput.

Consider alternatives for azeotropes or close-boilers or thermally sensitive feeds.

Batch mode for small/variable campaigns and Continuous mode for steady service.

Trays have easier cleaning, broad turndown and higher ΔP and packing needs good liquid distribution and has a low ΔP.

Pick pressure so condenser sees $\Delta T \geq 10–15$ °C and a mild vacuum for thermal stability.

Core Design

Sizing: Estimate Nmin (Fenske), Rmin (Underwood) and relate the N vs R (Gilliland).

Reflux target: R ≈ 1.2–1.7 x Rmin

- Typical binaries need around 11 to 25 theoretical stages.

Feed strategy: Match the q-line and locate near best K-value match. Avoid hydraulic bottlenecks

Hydraulics: Design at 70-85% of flood. The ΔP of tray must = 0.5-1.5 kPa per tray. The packing ΔP = 0.2-0.6 kPa/meter.

Heat Duties, Equipment, and Safety

Reboilers: Kettle (forgiving), thermosiphon (compact, NPSH and density-head)

Condensers: Total condenser for simplicity, use the partial to split light ends or control overhead composition.

Energy integration: connsider side reboilers/condensers, heat pumps as top-bottom ΔT is small.

- Exchanger- ΔT_{min} = 10°C and insulate the column.

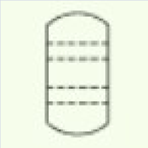
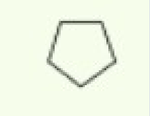
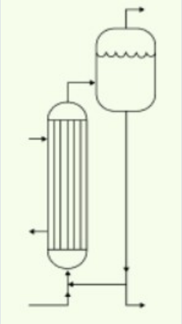
Materials for chlorides, use PSV for fire-utility loss and use vacuum breaker for sub-atm.

Control, Startup, and Troubleshooting

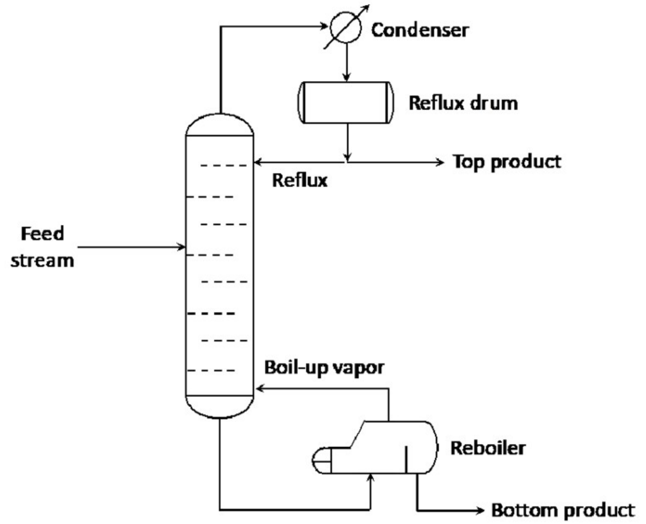
Control: level (bottoms/reflux drum), pressure via condenser

Fix one composition with reflux (D) and other with reboiler (B)

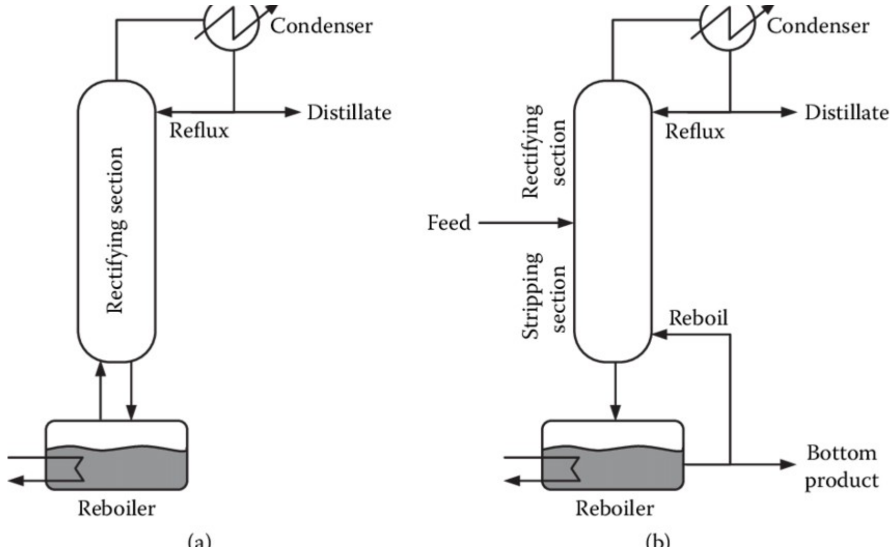
FloodingL

PFD Symbol [11]	Name
	Tray column
	Cyclone and Hydroclone
	Fluid separator
	Batch tray dryer
	Column
	Evaporator

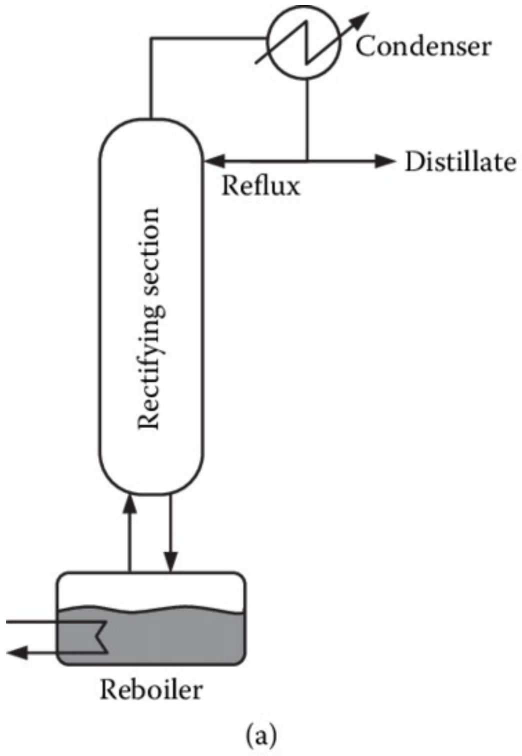
Distillation Columns



I. Batch
II. Continuous



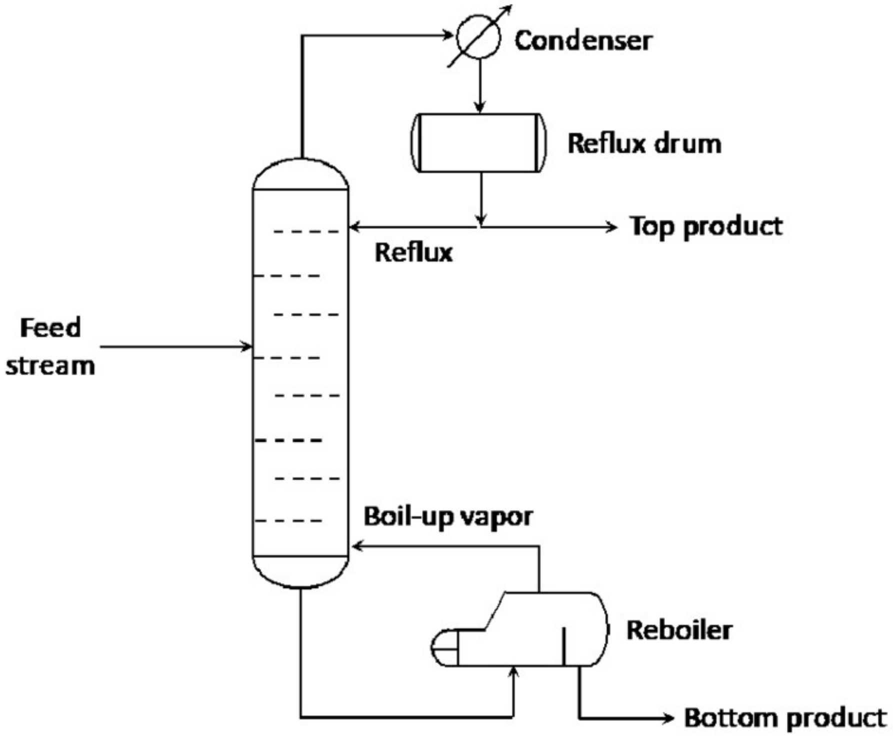
Batch Distillation



Schematic diagram of a Batch Distillation column. [10]

- Processes a fixed volume of mixture loaded into a still pot
- Mixture is heated; vapor rises through the column and separates components by volatility
- Composition in the pot changes over time → distillate composition also changes
- Operation continues until a target separation or final composition is reached
- After completion, the current batch is replaced and a new batch is started
- Ideal for small-scale operations and flexible production needs
- Common in specialty chemicals, pharmaceuticals, and research settings

Continuous Distillation

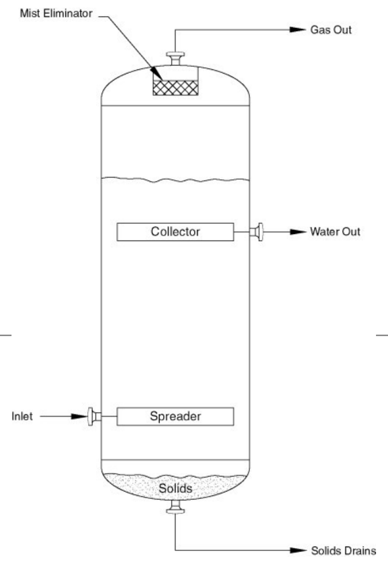


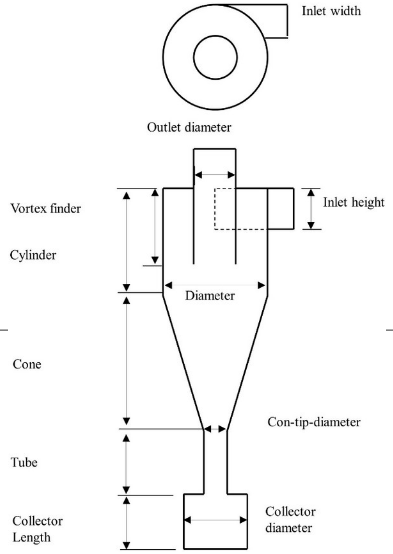
Schematic diagram of a continuous distillation column. [9]

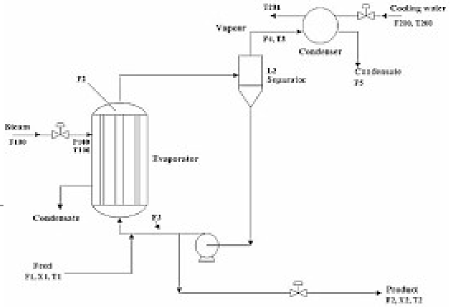
- Constant feed enters; distillate and bottoms are continuously withdrawn
- Heat supplied at reboiler and cooling at condenser to maintain operation
- Column reaches steady-state temperature, pressure, and composition profiles
- Produces consistent product purity over time
- Used for high-throughput, large-scale industrial separations
- Common in petroleum refining, petrochemicals, and bulk chemical production

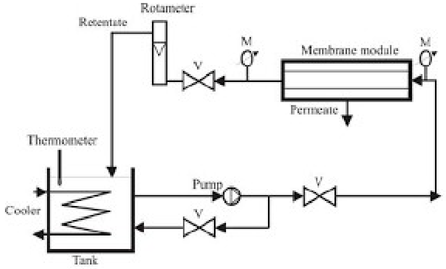
Separators

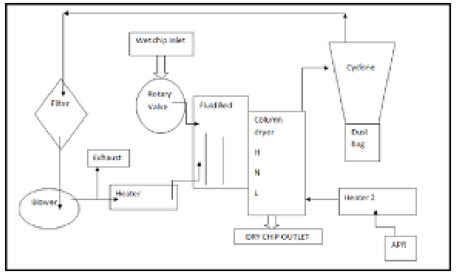
I. Mechanical/Gravity
II. Membrane
III. Thermal
IV. Adsorption

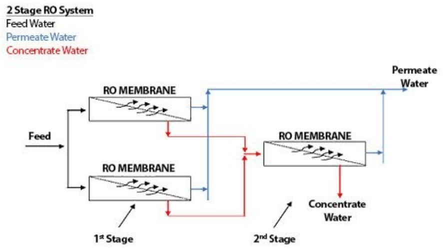


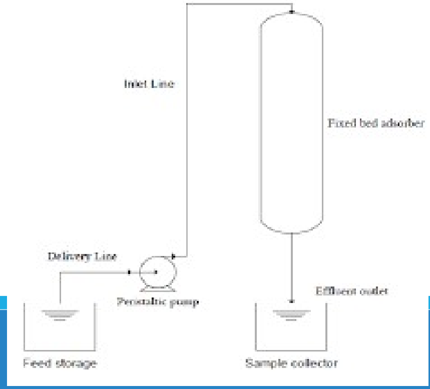


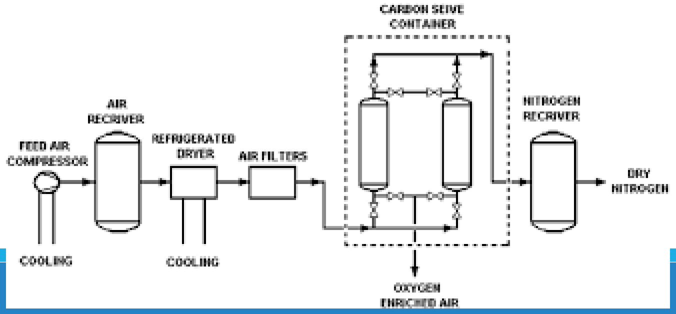


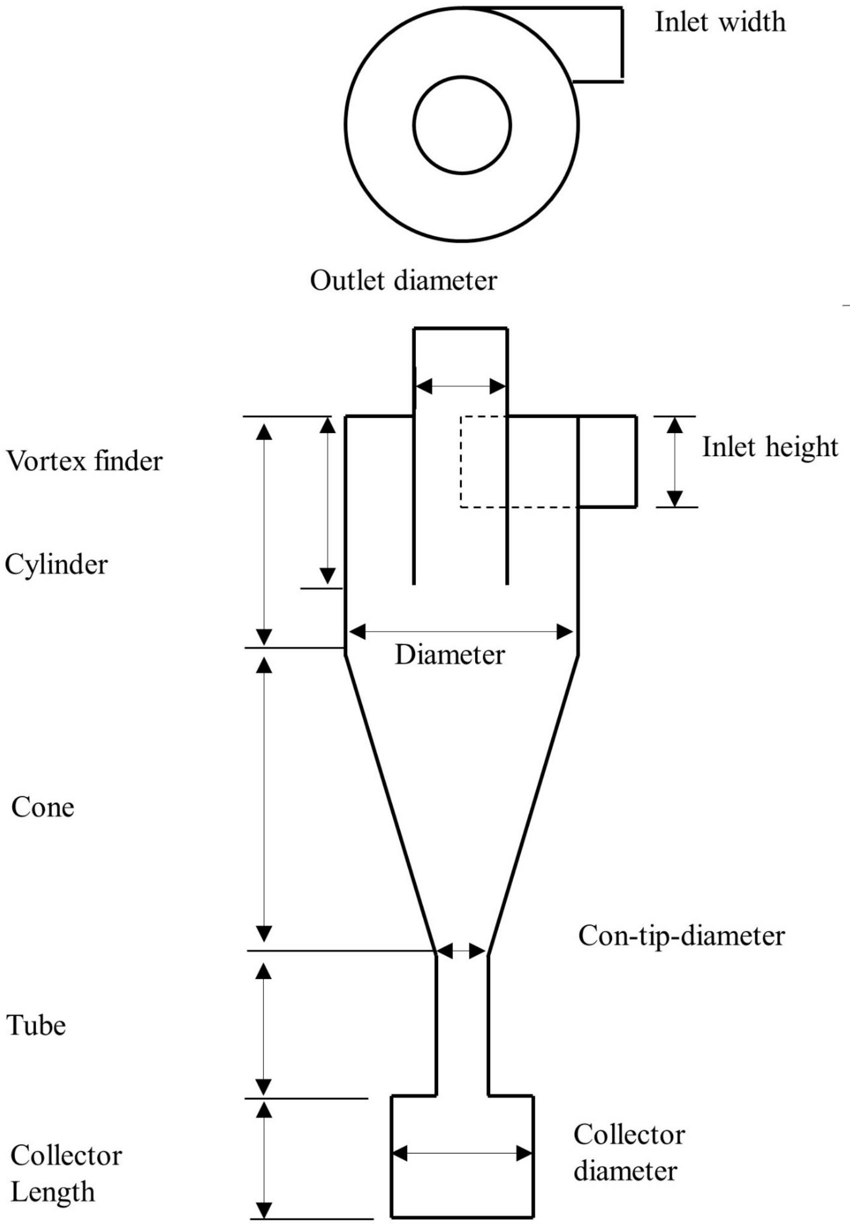








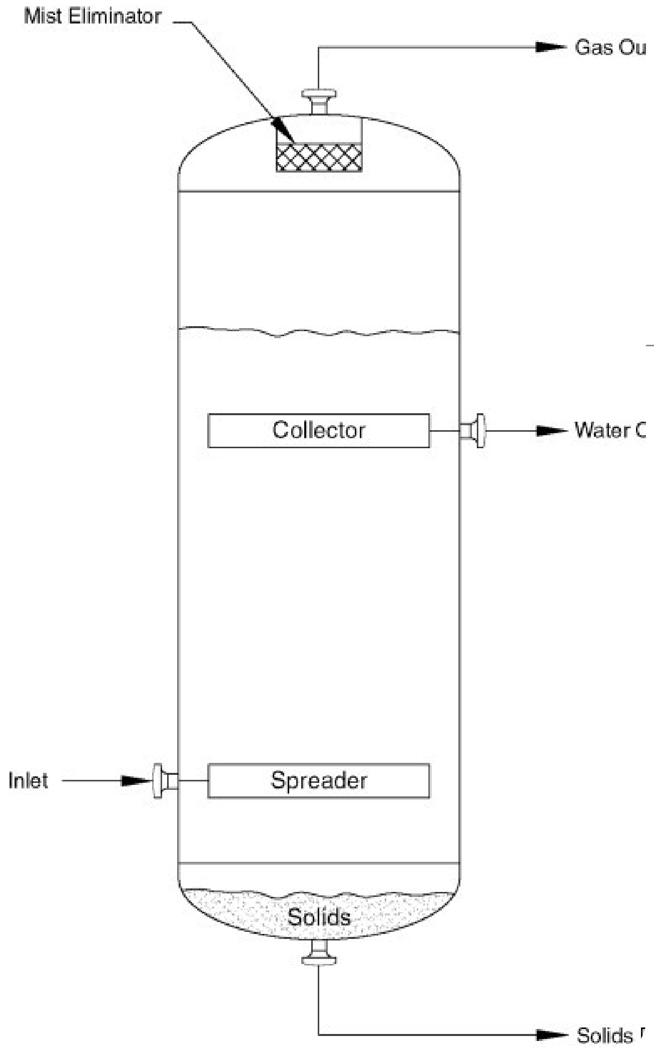




Schematic diagram of cyclone separator. [8]

Cyclone

- Uses centrifugal force to remove solids or droplets from a gas stream
- Gas enters tangentially, creating a spiral “tornado-like” motion
- Heavier particles are flung to the walls and fall to a collection chamber
- Clean gas exits upward through the center
- No filters required → so it is: simple, durable and low maintenance
- Used in chemical plants, power plants, woodworking and dust-collection systems

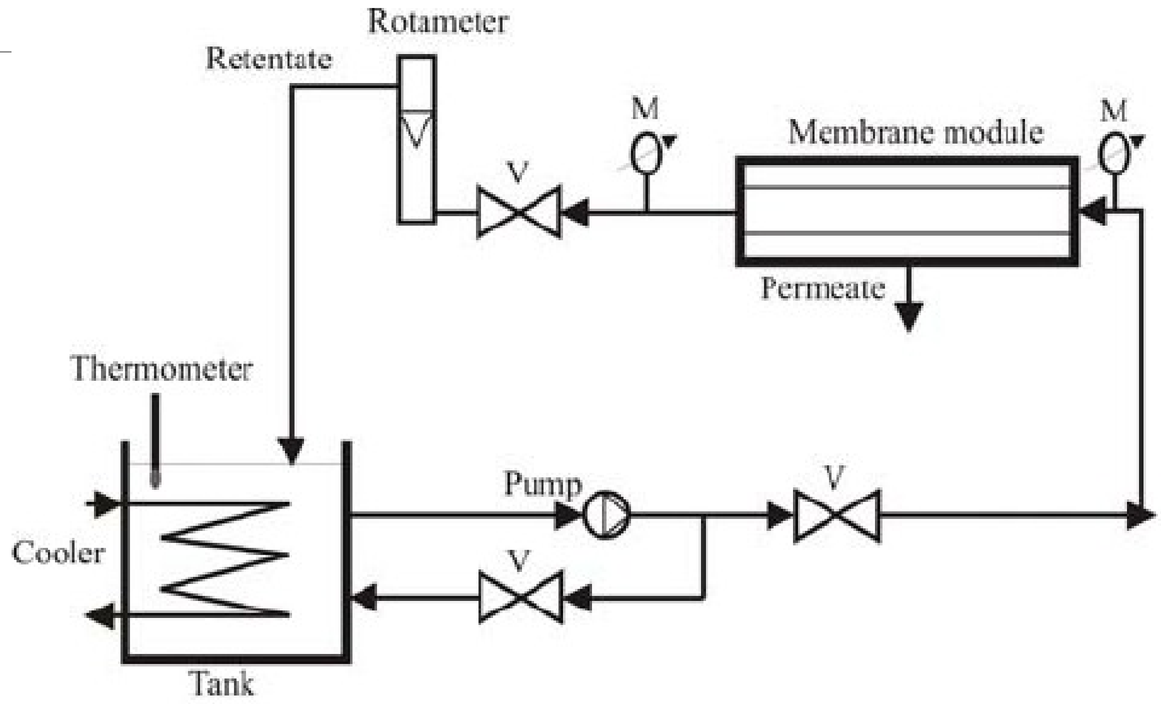


Gravity Settler

- Separates solids from liquids or two immiscible liquids using gravity
- Mixture enters a large, calm vessel allowing natural settling
- Heavier material sinks; lighter material rises or stays on top
- Distinct layers form and are removed separately
- Simple, low-energy separation method
- Common in wastewater treatment, mining, and slurry processing

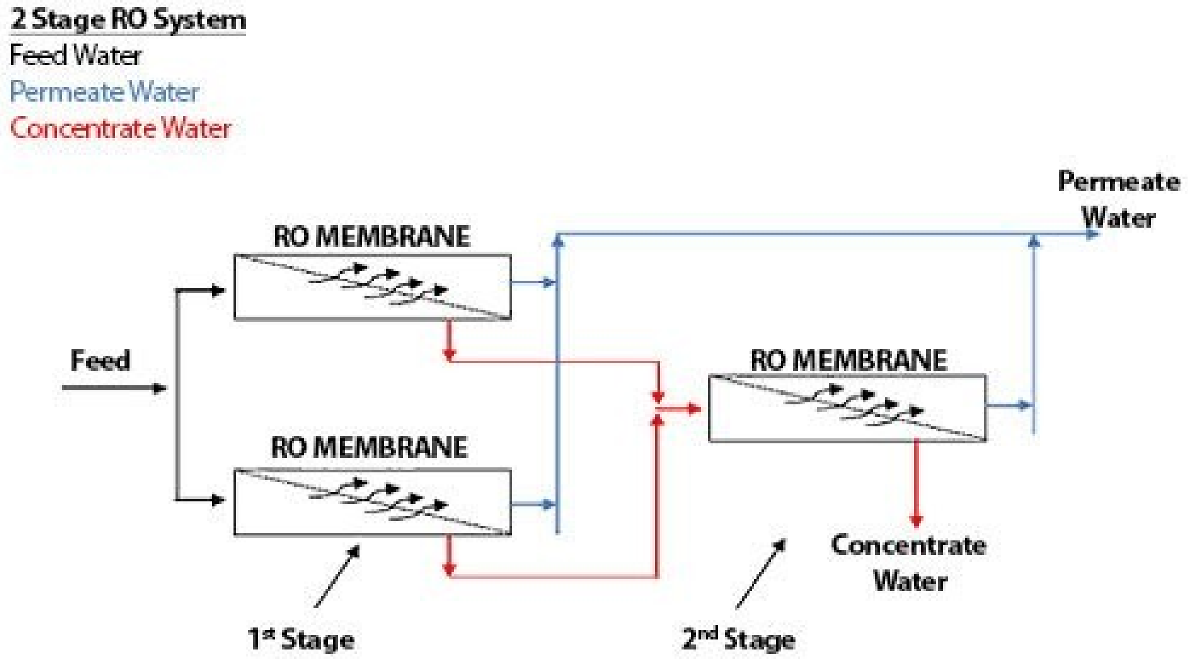
Membrane filters

- Uses a thin, selective barrier to separate components
- Only certain molecules or particles can pass through
- Smaller species permeate; larger contaminants are retained
- Works for liquids or gases without chemical additives or phase changes
- Provides fine, controlled separation
- Common in water purification, medical devices, and food processing



Membrane-filtration process apparatus where the membrane module (microfilter ceramic membranes) was installed. [6]

Reverse Osmosis

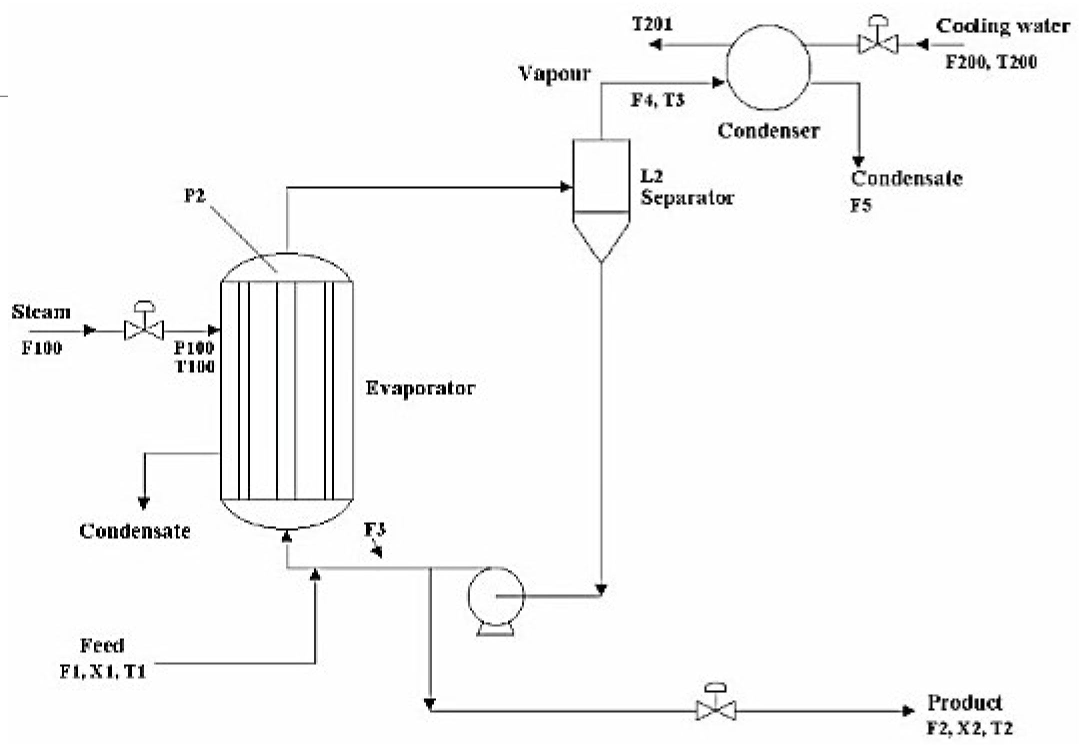


Schematic/process diagram of a reverse osmosis water-treatment system. [5]

- Uses a semipermeable membrane to remove dissolved salts and impurities
- Requires pressure greater than osmotic pressure to drive water through the membrane
- Water passes through; dissolved solids and contaminants are rejected as concentrate
- Produces high-purity water
- Commonly used for desalination, boiler feedwater, and pharmaceutical-grade water production

Evaporator

- Concentrates a solution by heating to vaporize and remove the solvent
- Leaves behind a more concentrated liquid product
- Used to increase solids content or recover valuable components
- Multiple-effect and vacuum designs improve energy efficiency
- Vacuum operation protects heat-sensitive materials
- Common in food, pharmaceutical, and chemical industries



Simplified process-flow diagram of evaporator system. [4]

Dryer

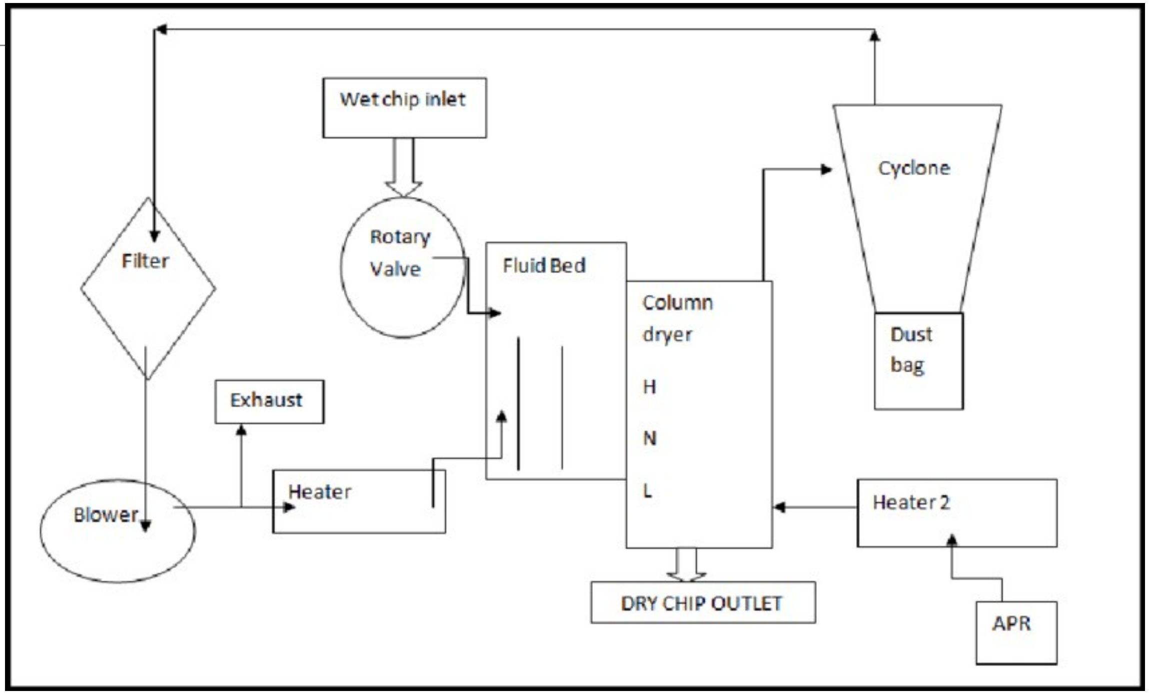
Removes moisture from solids or liquids using heat

Often combined with airflow or vacuum to evaporate and remove water/solvent

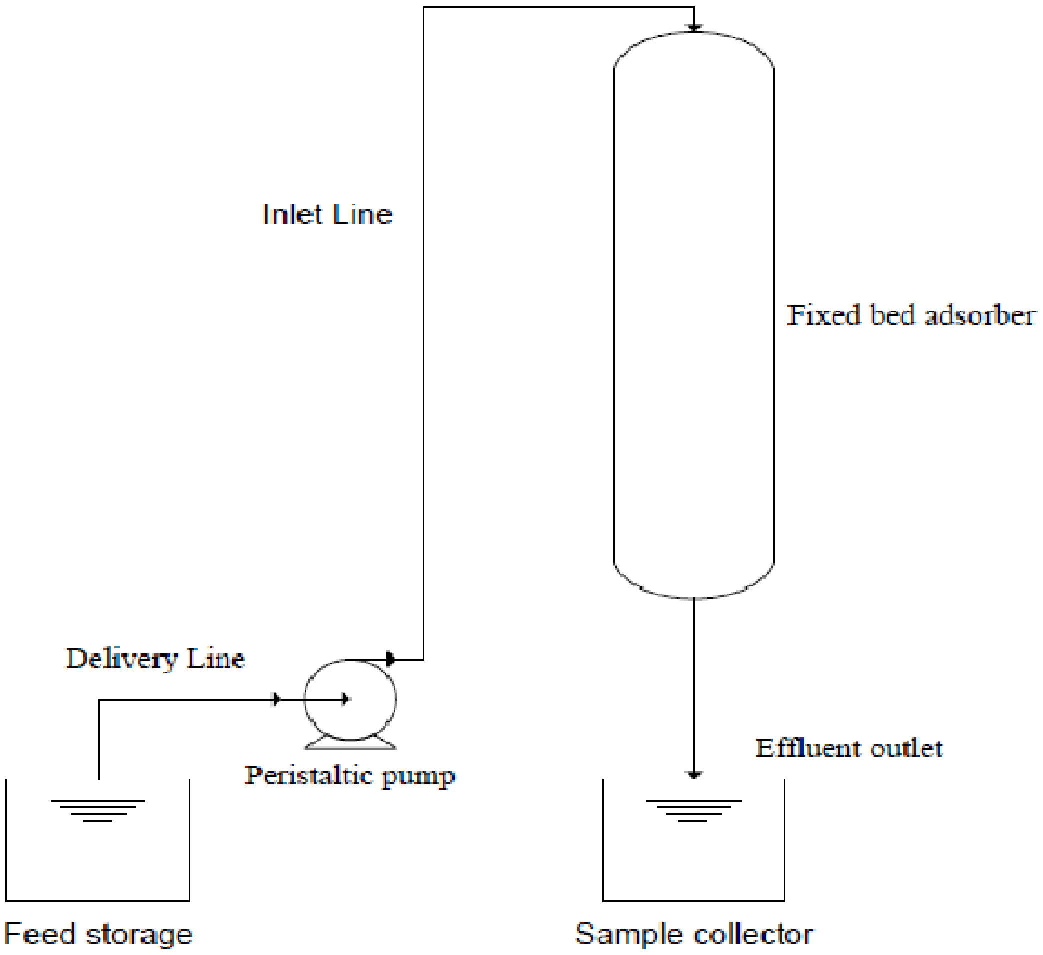
Produces dry, stable, and transportable products

Common designs include spray, rotary, and fluid-bed dryers

Widely used in food processing, pharmaceutical, and chemical industries



Process-flow diagram of the dryer. [3]



Schematic diagram of fixed-bed column adsorber. [2]

Fixed Bed Adsorber

- Gas or liquid flows through a packed bed of solid adsorbent
- Adsorbent selectively captures contaminants or target molecules
- Bed is regenerated or replaced when saturated
- Provides efficient removal of trace compounds
- Commonly used for gas purification, solvent recovery, and water treatment

Pressure Swing Adsorption System

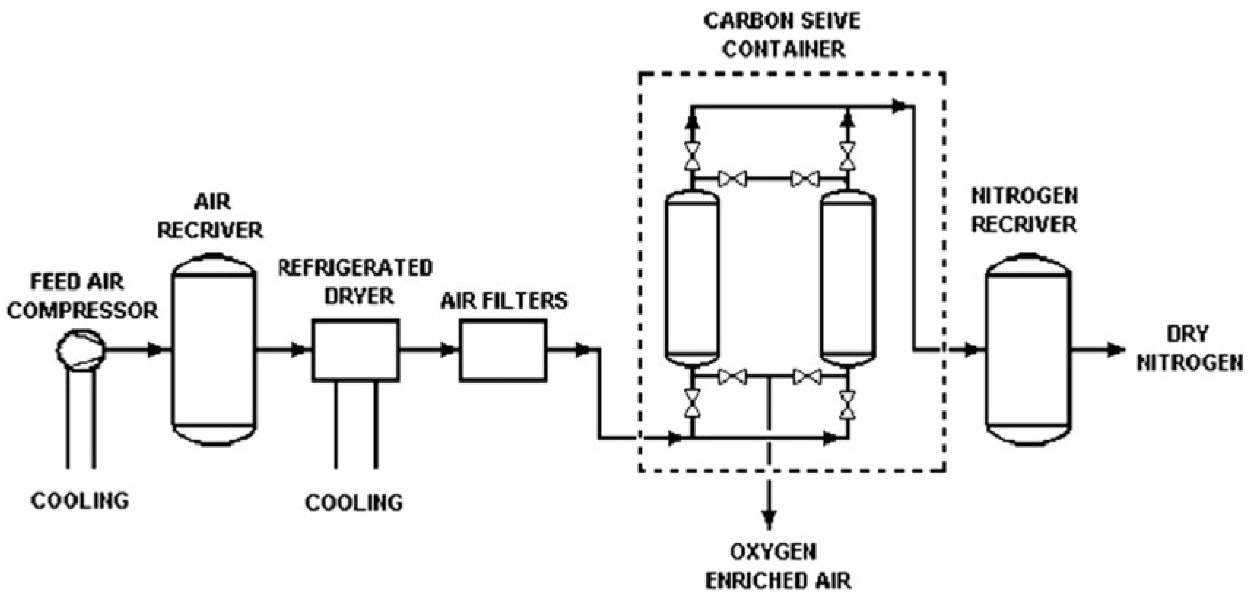
Separates gases using packed beds of adsorbent

Cycles between high and low pressure

At high pressure, target gases are adsorbed; others pass through as product

Low pressure releases adsorbed gases and regenerates the bed

Used for oxygen and nitrogen production, hydrogen purification, and refinery gas treatment



Schematic diagram of a pressure-swing adsorption system. [1]

Conclusion

There are a variety of different ways to separate mixtures, each with their own specific advantages and disadvantages.

Separation techniques frequently rely on the phases coming in and out of the separator - it's important to consider the phases components will be in as they go through a process.

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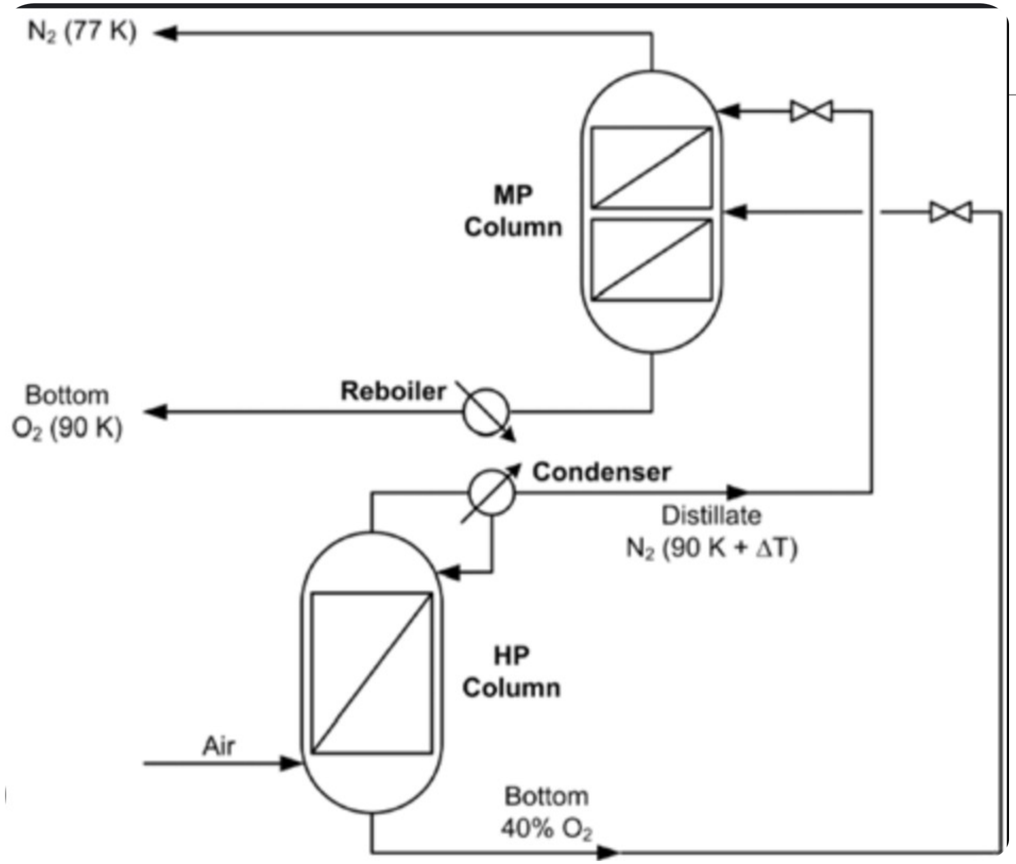
Cryogenic Distillation



Separation of Nitrogen/Oxygen and sometimes Argon

Relies on a pressure increase to the air feed to aid in the change of state from gas to liquid, then uses the low boiling points for a highly effective separation

A highly energy intensive process

Pressure Swing Distillation

Utilizes two different distillation columns at different pressures, with a recycle stream from the second column to the first

Good for separation azeotropes – relies heavily on azeotropic data

Cost benefits of being able to effectively use a heat exchanger within the continuous process